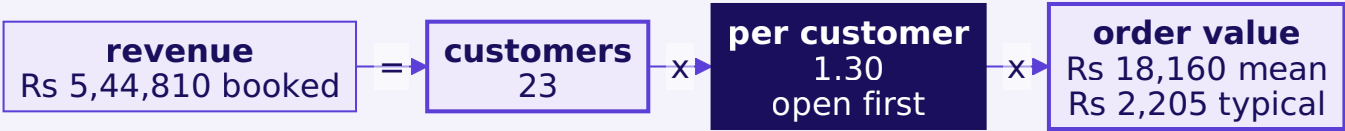


What is revenue made of, and which check catches each wrong number?

Kalpa Retail, Week 1 Monday. Before Meera Raghavan signs Rs 12 crore for new customers, is acquisition even the branch that is short? On 30 orders from 1 July to 26 September, each chapter met a plausible wrong number that one check caught before it reached her.

1 What is revenue made of, with Monday's numbers on it?



Crux. Revenue is customers times orders per customer times average order value; the identity $23 \times (30 / 23) \times (\text{Rs } 5,44,810 / 30)$ lands on Rs 5,44,810.

2 Chapter 1: which total is sales?

Wrong number	Rs 5,44,810 is called sales, with 4 cancelled store orders inside it.
Check	Count by status first: 21 delivered, 5 returned, 4 cancelled.
Fix	Each total named, with the bridge: booked Rs 5,44,810, less Rs 9,050 cancelled, less Rs 14,970 returned.

Crux. Name the definition before the number: booked, "not cancelled", or delivered.

5 Chapter 4: what is a typical order?

Wrong number	A first order is valued at the mean, Rs 18,160.
Check	Orders above the mean: 1 of 30.
Fix	The median, Rs 2,205 booked, is the typical order; the mean stays for totals.

Crux. Report the median when one order can move the mean, and say why.

7 Chapter 6: are the one-time buyers lost?

Wrong number	16 of 23 bought once, so "70 percent are lost".
Check	Median repeat gap 45 days; 9 of the 16 bought in the last 45.
Fix	7 came back, 7 are past the usual gap, 9 are too recent to judge.

Crux. A one-time buyer is not a lost customer until they have had time to come back.

3 Chapter 2: why must a branch use one definition?

Wrong number	AOV is Rs 25,943: booked rupees over delivered orders.
Check	AOV x the 30 orders the revenue covers gives Rs 7,78,300 against Rs 5,44,810.
Fix	Rs 18,160 booked, or Rs 24,800 delivered, each named.

Crux. Build every fraction on one definition, and check that it multiplies back.

6 Chapter 5: which branch first?

Wrong number	Customers +10 percent and frequency +10 percent called 20 percent, Rs 77,772.
Check	$1.10 \times 1.10 = 1.21$: Rs 78,420 on the consumer segments' Rs 64,810.
Fix	Rs 9,722 more needs 15 percent more customers or orders; frequency has the evidence.

Crux. Lifts multiply along the tree: two 10 percent lifts make 21 percent.

8 What will Meera sign?

"On 30 booked orders, 23 customers placed 1.30 orders each at a typical Rs 2,205; 7 came back and 9 are too recent to judge, so open frequency first and hold the Rs 12 crore for Tuesday's two quarters."

Crux. The sentence names its definition, its branch and its caveat, and waits for a second quarter before the budget.

4 Chapter 3: do customers come back?

Wrong number	30 customers, so $30 / 30 = 1.00$ and "nobody comes back".
Check	<code>len(rows)</code> against <code>len(set(ids))</code> : 30 against 23.
Fix	23 customers, 1.30 orders each; 7 came back and 16 bought once.

Crux. Count customers by their id, never by the rows.

THE WORDS ON THIS SHEET

Revenue tree Revenue is split into metrics that multiply.

Booked revenue It counts every order placed.

Delivered revenue It counts orders that arrived and stayed.

Denominator It is what a metric divides by.

Identity check Branches multiplied back must give the total.

Distinct customers Each customer id is counted once.